

Multivariate OLS Regression: Predicting Titanic Fare

Text written in black is generated by Claude, text in red is written by me.

1 Model Setup

Let n be the number of passengers in the sample. Define

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \quad (1)$$

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ \vdots & \vdots & \vdots \\ x_{n1} & x_{n2} & x_{n3} \end{bmatrix} \in \mathbb{R}^{n \times 3}, \quad (2)$$

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \in \mathbb{R}^3 \quad (3)$$

where y_i is Fare for passenger i , and row i of $\mathbf{X}$ is $(\text{Pclass}_i, \text{Age}_i, \text{SibSp}_i)$. I haven't given the model the dataset description, which is why it seems like it has no idea what these variables mean. `pclass` is the ticket class, `sibsp` is the number of siblings/spouses aboard the Titanic, and well, `Age` is age.)

The model is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \in \mathbb{R}^n \quad (4)$$

or, elementwise, for passenger i :

$$y_i = \beta_1 \text{Pclass}_i + \beta_2 \text{Age}_i + \beta_3 \text{SibSp}_i + \varepsilon_i \quad (5)$$

2 OLS Loss Function

$$L(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 \quad (6)$$

This looks good, we are setting up an ordinary least squares function where we subtract the linear regression function of $\mathbf{X}$ s and $\boldsymbol{\beta}$ s from the linear model $\mathbf{Y}$. For my own understanding: suppose we have a vector

$\mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$. Multiplying $\mathbf{e}^\top \mathbf{e} = [e_1 \ e_2 \ e_3] \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = e_1^2 + e_2^2 + e_3^2$ so we should think $\mathbf{v}^\top \mathbf{v} = |\mathbf{v}|^2 = \sum_i v_i^2$. In

the case of regression, we have $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ (difference between data points and the linearly predicted value), where $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta} \implies \mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta} \implies \mathbf{e}^\top \mathbf{e} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$. We are aiming to minimize this through differentiation as shown below.

3 First Derivative

Expanding the quadratic form:

$$L(\beta) = \mathbf{Y}^\top \mathbf{Y} - \mathbf{Y}^\top \mathbf{X} \beta - \beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta \quad (7)$$

Since $\mathbf{Y}^\top \mathbf{X} \beta$ is a scalar, it equals its own transpose:

$$\mathbf{Y}^\top \mathbf{X} \beta = \beta^\top \mathbf{X}^\top \mathbf{Y} \quad (8)$$

so

$$L(\beta) = \mathbf{Y}^\top \mathbf{Y} - 2\beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta \quad (9)$$

Using the matrix-calculus identities **WHY?? check matrix differentiation**

$$\frac{\partial(\mathbf{a}^\top \beta)}{\partial \beta} = \mathbf{a}, \quad \frac{\partial(\beta^\top \mathbf{A} \beta)}{\partial \beta} = 2\mathbf{A} \beta \quad (\text{for symmetric } \mathbf{A}) \quad (10)$$

we differentiate term by term:

$$\frac{\partial L}{\partial \beta} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X} \beta \quad (11)$$

4 Setting the Gradient to Zero

$$-2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X} \beta = 0 \implies \mathbf{X}^\top \mathbf{X} \beta = \mathbf{X}^\top \mathbf{Y} \quad (12)$$

These are the *normal equations*. Assuming $\mathbf{X}^\top \mathbf{X}$ is invertible:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \quad (13)$$

This is a minimum because the Hessian

$$\frac{\partial^2 L}{\partial \beta \partial \beta^\top} = 2\mathbf{X}^\top \mathbf{X} \quad (14)$$

is positive semi-definite (positive definite whenever $\mathbf{X}$ has full column rank).

5 A Concrete 3×3 Example from the Data

Taking the first three complete rows of the Titanic training data (Pclass, Age, SibSp $\rightarrow$ Fare):

$$\mathbf{X} = \begin{bmatrix} 3 & 22 & 1 \\ 1 & 38 & 1 \\ 3 & 26 & 0 \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} 7.25 \\ 71.2833 \\ 7.925 \end{bmatrix} \quad (15)$$

6 Numerical Solution and Gradient at that Point

Since $\mathbf{X}$ is square, this is an exactly-determined system rather than an over-determined least-squares problem:

$$\det(\mathbf{X}) = -100 \neq 0 \implies \mathbf{X}^{-1} \text{ exists} \quad (16)$$

$$\mathbf{X}^{-1} = \begin{bmatrix} 0.26 & -0.26 & 0.16 \\ -0.03 & 0.03 & 0.02 \\ 0.88 & 0.12 & -0.92 \end{bmatrix} \quad (17)$$

$$\hat{\beta} = \mathbf{X}^{-1}\mathbf{Y} = \begin{bmatrix} -15.38 \\ 2.08 \\ 7.64 \end{bmatrix} \quad (18)$$

Check:

$$\mathbf{X}\hat{\beta} = (7.25, 71.2833, 7.925)^\top = \mathbf{Y} \quad (19)$$

Gradient at $\hat{\beta}$:

$$\nabla L(\hat{\beta}) = -2\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\hat{\beta}) = -2\mathbf{X}^\top \cdot \mathbf{0} = \mathbf{0} \quad (20)$$

With exactly 3 data points and 3 parameters, $\mathbf{X}$ is (generically) invertible, so OLS degenerates into exact interpolation: the residual vector is forced to $\mathbf{0}$, and since the gradient is a linear function of the residual, it vanishes too.

7 Geometric Interpretation

1. **$\mathbf{X}$ as a linear map.** Think of $\mathbf{X} : \mathbb{R}^3 \rightarrow \mathbb{R}^n$ (parameter space to observation space). Each choice of β produces a predicted vector $\mathbf{X}\beta$, which lives in the *column space* of $\mathbf{X}$ — the span of the three predictor vectors (Pclass, Age, SibSp columns).
2. **$\mathbf{Y}$ generally isn't in that column space** when $n > p$, since real Fare data won't be an exact linear combination of the three columns — there's noise. We can't solve $\mathbf{X}\beta = \mathbf{Y}$ exactly, so we look for the closest achievable point.
3. **OLS = orthogonal projection.** Minimizing $\|\mathbf{Y} - \mathbf{X}\beta\|^2$ asks for the point in $\text{col}(\mathbf{X})$ closest to $\mathbf{Y}$ in Euclidean distance — the orthogonal projection of $\mathbf{Y}$ onto $\text{col}(\mathbf{X})$.
4. **The normal equations encode orthogonality.**

$$\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\beta) = \mathbf{0} \quad (21)$$

says the residual vector $\mathbf{Y} - \mathbf{X}\hat{\beta}$ is orthogonal to every column of $\mathbf{X}$, i.e., to the entire column space — the defining property of an orthogonal projection.

5. **The square-matrix case.** When $\mathbf{X}$ is square and invertible, $\text{col}(\mathbf{X}) = \mathbb{R}^3$, so every possible $\mathbf{Y}$ already lies in it. The “projection” of $\mathbf{Y}$ is just $\mathbf{Y}$ itself, the residual collapses to $\mathbf{0}$, and $\hat{\beta} = \mathbf{X}^{-1}\mathbf{Y}$ is a coordinate change rather than a genuine projection — consistent with the gradient vanishing trivially in Section 6.